A LOCALLY FINITE TYPE MORPHISM OF ALGEBRAIC SPACES IS LOCALLY QUASI-FINITE EXACTLY WHEN ITS FIELD-VALUED FIBRES ARE DISCRETE

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Editorial note. All delivered mathematical text is the author's own, copied from the pinned Stacks Project chapter file spaces-morphisms.tex; no statement or proof step was rewritten, reordered or removed.

Editorial note. The proof is the author's own two-way argument; the converse direction builds the contradiction from the points of the etale equivalence relation R_v rather than citing the scheme case.

Editorial note. The seven references of the counted spans are recorded as non-load-bearing navigation pointers; the argument that consumes them stays inside the counted proof span.

Editorial note. Preamble: the three native excerpt spans of sources/stacks-spaces-morphisms/preamble.tex (lines 8-13, 14-18 and 154-217). The chapter file itself carries no preamble text.

Allowed standard prerequisites (editorial).

- Base change of etale morphisms of algebraic spaces. Properties of Spaces, Lemma spaces-properties-lemma-base-change-etale, used at proof lines 5420-5422. Makes U_k -> X_k etale, hence open and surjective on points.
- Stability of local quasi-finiteness under base change. Lemma lemma-base-change-quasi-finite, used at proof lines 5423-5425. Transfers local quasi-finiteness to the fibre X_k.
- The scheme criterion: local quasi-finiteness at a point via discrete fibres. Morphisms, Lemmas morphisms-lemma-locally-quasi-finite-fibres and morphisms-lemma-quasi-finite-at-point-characterize, used at proof lines 5427-5428 and 5444-5447. Gives discreteness in one direction and produces the specialization in the other.
- Dimension formula for specialisations of points of a scheme over a field. Morphisms, Lemma morphisms-lemma-dimension-fibre-specialization, used at proof lines 5451-5454. Yields the strict inequality of transcendence degrees.
- Residue fields of etale morphisms over a field are finite separable extensions. Morphisms, Lemma morphisms-lemma-etale-over-field, and Properties of Spaces, Lemma spaces-properties-lemma-points-cartesian, used at proof lines 5458-5464. Forces the two transcendence degrees to be equal.

Source reference. Original lemma 03XL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 5152–5152. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 06RU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8653–8653. Navigation pointer into the Stacks

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Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 02GL: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 3232–3232. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 06RT: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 8652–8652. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 01TH: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 2374–2374. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03FU: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4505–4505. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Source reference. Original lemma 03H4: focused-seeds/AGP/sources/stacks-snapshot-20261011/tags.txt, source lines 4553–4553. Navigation pointer into the Stacks Project; the standard fact it supplies is applied inside the counted proof span and the argument resting on it stays there.

Lemma 0.2. *Let S be a scheme. Let $f : X \to Y$ be a morphism of algebraic spaces. Assume f is locally of finite type. The following are equivalent*

1. *f is locally quasi-finite,*
2. *for every morphism $\mathrm{Spec}(k) \to Y$ where k is a field the space $|X_k|$ is discrete. Here $X_k = \mathrm{Spec}(k) \times_Y X$.*

Proof. Assume f is locally quasi-finite. Let $\mathrm{Spec}(k) \to Y$ be as in (2). Choose a surjective étale morphism $U \to X$ where U is a scheme. Then $U_k = \mathrm{Spec}(k) \times_Y U \to X_k$ is an étale morphism of algebraic spaces by Properties of Spaces, Lemma 03FU. By Lemma 03XL we see that $X_k \to \mathrm{Spec}(k)$ is locally quasi-finite. By definition this means that $U_k \to \mathrm{Spec}(k)$ is locally quasi-finite. Hence $|U_k|$ is discrete by Morphisms, Lemma 06RT. Since $|U_k| \to |X_k|$ is surjective and open we conclude that $|X_k|$ is discrete.

Conversely, assume (2). Choose a surjective étale morphism $V \to Y$ where V is a scheme. Choose a surjective étale morphism $U \to V \times_Y X$ where U is a scheme. Note that $U \to V$ is locally of finite type as f is locally of finite type. Picture

$$\begin{array}{ccccc} U & \longrightarrow & X \times_Y V & \longrightarrow & V \\ & \searrow & \downarrow & & \downarrow \\ & & X & \longrightarrow & Y \end{array}$$

If f is not locally quasi-finite then $U \to V$ is not locally quasi-finite. Hence there exists a specialization $u \rightsquigarrow u'$ for some $u, u' \in U$ lying over the same point $v \in V$, see Morphisms, Lemma 01TH. We claim that u, u' do not have the same image in $X_v = \mathrm{Spec}(\kappa(v)) \times_Y X$ which will contradict the assumption that $|X_v|$ is discrete as desired. Let $d = \mathrm{trdeg}_{\kappa(v)}(\kappa(u))$ and $d' = \mathrm{trdeg}_{\kappa(v)}(\kappa(u'))$. Then we see that $d > d'$ by Morphisms, Lemma 06RU. Note that U_v (the fibre of $U \to V$ over v) is the fibre product of U and X_v over $X \times_Y V$, hence $U_v \to X_v$ is étale (as a base change of the

étale morphism $U \to X \times_Y V$). If $u, u' \in U_v$ map to the same element of $|X_v|$ then there exists a point $r \in R_v = U_v \times_{X_v} U_v$ with $t(r) = u$ and $s(r) = u'$, see Properties of Spaces, Lemma 03H4. Note that $s, t : R_v \to U_v$ are étale morphisms of schemes over $\kappa(v)$, hence $\kappa(u) \subset \kappa(r) \supset \kappa(u')$ are finite separable extensions of fields over $\kappa(v)$ (see Morphisms, Lemma 02GL). We conclude that the transcendence degrees are equal. This contradiction finishes the proof. $\square$